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Event-Contextual Geopolitical Risk and Brent Oil Prices: A Comparison of the Russia–Ukraine War and Iran Tensions Using an Interaction-Augmented NARDL Approach

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Event-Contextual Geopolitical Risk and Brent Oil Prices: A Comparison of the Russia–Ukraine War and Iran Tensions Using an Interaction-Augmented NARDL Approach

Abstract

This study examines whether the impact of geopolitical risk on Brent oil prices varies across event contexts. Using an interaction-augmented NARDL approach, the study shows that geopolitical risk should not be treated merely as a general source of uncertainty, but as a risk factor priced through the geographical position of risk-producing actors and their location within the energy supply chain. Interpreted cautiously, the findings suggest that the Russia–Ukraine War generated a stronger geopolitical risk transmission to Brent oil prices than Iran-related tensions. Overall, the results indicate that geopolitical risk is not a homogeneous shock in oil markets, but a spatially embedded and context-specific pricing mechanism.

Keywords: Geopolitical risk; Brent oil prices; Russia–Ukraine War; Iran-related tensions; NARDL; Energy security

1. Introduction

Crude oil prices are among the key indicators reflecting the functioning of the global energy system, expectations regarding supply security, and perceptions of geopolitical risk. These prices do not merely respond to current supply–demand conditions; they also price in expectations of future supply disruptions, transportation risks, and the likelihood of conflict (Kilian et al., 2024). Therefore, fluctuations in oil prices have direct implications for energy costs, inflation dynamics, fiscal balances, market expectations, and the policy choices of developing economies, particularly those dependent on energy imports (Yang et al., 2023; Huang et al., 2021).

Given that states often regard oil as an important political instrument, oil markets are inevitably exposed to geopolitical events and tensions (Qin et al., 2020). This exposure may operate through two opposing channels. On the one hand, geopolitical tensions can raise the probability of future supply disruptions and generate a risk premium in oil markets. On the other hand, they can increase uncertainty about global economic activity, thereby exerting downward pressure on consumption, investment, and energy demand (Kilian et al., 2024; Minesso et al., 2023). Accordingly, the impact of geopolitical risk on oil prices may vary not only with the level of risk itself, but also with the geography of the event, the affected supply or transit channel, and the expectations through which the market prices the shock.

The existing literature shows that the relationship between geopolitical risk and oil prices is neither linear, unidirectional, nor stable over time. Studies based on the GPR index demonstrate that geopolitical risk can affect oil prices, oil returns, volatility, and inflation dynamics (Caldara and Iacoviello, 2022; Yang et al., 2023; Ivanovski and Hailemariam, 2022; Qin et al., 2020). However, much of this literature treats geopolitical risk as a general uncertainty indicator and only partially distinguishes the channels through which different geopolitical events are transmitted to oil prices. Yet the effect of geopolitical risk on oil markets may depend not only on the level of risk, but also on whether the risk emerges as a producer-country crisis, a sanctions regime, a transit-security threat, or a vulnerability at a critical energy chokepoint.

In this context, the Russia–Ukraine War and Iran-related tensions can be considered two distinct types of geopolitical risk transmitted to oil markets through different channels. The Russia–Ukraine War represents a geopolitical shock that directly affects an energy-producing and exporting region, shaping oil prices through sanctions, supply expectations, European energy security concerns, and broader market uncertainty. By contrast, Iran-related tensions are priced less through actual oil supply losses and more through transit security, tanker transportation, insurance costs, and expectations of potential supply disruptions, particularly because of the strategic position of the Strait of Hormuz. Given the central role of the Strait of Hormuz in global oil flows, Iran-related tensions may influence oil markets primarily through the oil risk premium channel (Olanipekun and Alola, 2020; Shepard and Pratson, 2020).

This study builds on this distinction by examining whether the impact of geopolitical risk on Brent oil prices varies across event contexts. To this end, the Global Geopolitical Risk Index is decomposed into positive and negative components, which are then interacted with dummy variables representing the Russia–Ukraine War and Iran-related tensions. This design allows the study to evaluate, within a single empirical framework, both the general effect of geopolitical risk on Brent prices and the extent to which this effect changes across two distinct geopolitical contexts.

2. Literature Review and Conceptual Background

2.1. Geopolitical risk and oil price dynamics

The Global Geopolitical Risk Index developed by Caldara and Iacoviello (2022) is one of the most widely used indicators for measuring geopolitical risk. Based on systematic newspaper searches, the index captures not only realized geopolitical events but also threats, tensions, and expectations of conflict. This makes the GPR index particularly useful for analysing oil markets, where risk is often priced before actual supply disruptions materialize.

Recent studies show that geopolitical risk affects oil markets through multiple and sometimes opposing channels. Yang et al. (2023) emphasize two mechanisms: an economic activity channel, through which geopolitical uncertainty weakens consumption, investment, and oil demand; and a risk channel, through which expectations of future supply disruptions raise the holding value of oil contracts and increase risk premiums. Similarly, the ECB assessment summarized by Minesso et al. (2023) shows that the oil price response to geopolitical shocks depends on the source of the event and the market's perception of supply disruption risks.

The literature also indicates that the GPR–oil price nexus is time-varying and conditional on market regimes. Ivanovski and Hailemariam (2022) show that the relationship between geopolitical risk, oil prices, and oil price volatility changes over time, while Huang et al. (2021) find that geopolitical risk is more strongly associated with oil volatility than with oil returns. These findings support the view that geopolitical risk should not be treated as a uniform shock in oil markets.

2.2. Nonlinear and asymmetric effects of geopolitical risk

Because geopolitical risk may affect oil prices differently depending on the direction of the shock and market conditions, nonlinear and asymmetric approaches have become increasingly important. Qin et al. (2020) show that the effects of geopolitical risk on energy returns and volatility vary across market conditions, while Ren et al. (2023), using a QARDL approach, demonstrate that the effect of GPR on crude oil prices differs across quantiles and time horizons. These studies suggest that the oil market response to geopolitical risk is not adequately captured by linear and symmetric specifications.

Asymmetric modelling is particularly relevant for oil markets because increases in geopolitical risk may generate immediate risk premiums, whereas reductions in risk may not lead to proportional price declines. Wars, sanctions, and regional tensions may leave persistent uncertainty in expectations even after the initial shock fades. The NARDL framework is therefore suitable for separating positive and negative changes in geopolitical risk and for evaluating whether their effects differ in the short and long run (Shin et al., 2014; Pesaran et al., 2001).

2.3. Event-contextual geopolitical risk: Russia–Ukraine and Iran-related tensions

The Russia–Ukraine War represents a producer-country and sanctions-related geopolitical shock. Zhang et al. (2024) show that the war and associated sanctions generated sharp increases in WTI and Brent prices and substantially increased market volatility. Fang and Shao (2022), Ohikhuare (2023), and Pan and Sun (2023) further show that the war affected commodity markets through volatility, financial spillovers, futures market reactions, and broader uncertainty channels. These findings indicate that the Russia–Ukraine War is not merely a general increase in geopolitical risk, but an event context linked to production geography, sanctions, and European energy security.

Iran-related tensions represent a different geopolitical pricing mechanism. Rather than directly affecting a major production area in the same way as the Russia–Ukraine War, Iran-related tensions are primarily linked to the Strait of Hormuz, tanker transportation, transit security, insurance costs, and expectations of possible supply disruptions. Shepard and Pratson (2020) identify the Strait of Hormuz as a critical chokepoint for global energy flows, while Goldenberg et al. (2019) emphasize that US–Iran tensions can be priced through low-probability but high-impact scenarios involving shipping traffic and energy infrastructure. Olanipekun and Alola (2020) also show that geopolitical risk in the Persian Gulf can affect oil production asymmetrically.

Overall, the literature reveals three gaps. First, many studies examine the oil price effects of geopolitical risk through the aggregate GPR index without testing whether the effect varies across event contexts. Second, the Russia–Ukraine War and Iran-related tensions are usually discussed in separate literatures, and their effects on Brent prices are rarely compared within the same empirical framework. Third, existing studies focus mostly on direct price, volatility, or spillover effects, whereas fewer studies model GPR components through event-specific interaction terms. This study addresses these gaps by comparing two geopolitical event contexts within an interaction-augmented NARDL framework.

3. Data and Empirical Strategy

3.1. Data and variables

This study examines the relationship between Brent oil prices and geopolitical risk using monthly data for the period 2007M06–2026M03. The dataset is designed to test both the general effect of geopolitical risk on Brent oil prices and whether this effect differs across the Russia–Ukraine War and Iran-related tension contexts. The main variables are Brent oil prices, the Global Geopolitical Risk Index, Russia–Ukraine and Iran event dummies, and the OVX and EPUTRADE variables used in robustness checks.

Table 1. Variable definitions and data sources

Variable	Symbol	Description	Source	Frequency
Brent oil price	BRENT	Brent crude oil price	Investing.com / EIA / FRED	Monthly
Global Geopolitical Risk Index	GPR	News-based global geopolitical risk index	Caldara and Iacoviello	Monthly
Russia–Ukraine War dummy	RU_D	Event dummy representing the Russia–Ukraine War period	Author’s definition	Monthly
Iran-related tension dummy	IR_D	Event dummy representing the Iran–US–Israel tension period	Author’s definition	Monthly
Crude Oil Volatility Index	OVX	Indicator of oil market volatility and risk perception	CBOE / FRED	Monthly
Trade Policy Uncertainty Index	EPUTRADE	Index representing trade policy uncertainty	FRED	Monthly

3.2. Variable construction

BRENT, GPR, OVX, and EPUTRADE are used in logarithmic form. Following the NARDL approach, the logarithmic GPR series is decomposed into positive and negative partial sums. The event-specific interaction terms are then constructed by multiplying these GPR components with the Russia–Ukraine and Iran event dummies:

$$GPR_POS(t) = \sum[j = 1, \dots, t] \max(\Delta \ln GPR(j), 0)$$

$$GPR_NEG(t) = \sum[j = 1, \dots, t] \min(\Delta \ln GPR(j), 0)$$

$$GPR_POS_RU(t) = GPR_POS(t) \times RU_D(t)$$

$$GPR_NEG_RU(t) = GPR_NEG(t) \times RU_D(t)$$

$$GPR_POS_IR(t) = GPR_POS(t) \times IR_D(t)$$

$$GPR_NEG_IR(t) = GPR_NEG(t) \times IR_D(t)$$

These interaction terms are included to test whether the effect of geopolitical risk on Brent oil prices changes across event contexts. OVX and EPUTRADE are used in robustness analyses to assess whether the main findings are preserved after controlling for oil market volatility and trade policy uncertainty.

3.3. Event-contextual NARDL specification

The event-contextual long-run relationship is defined through the following functional form:

$$\ln BREN T(t) = f[GPR_POS(t), GPR_NEG(t), GPR_POS_RU(t), GPR_NEG_RU(t), GPR_POS_IR(t), GPR_NEG_IR(t)]$$

Based on this functional structure, the NARDL model is estimated in an error-correction form that jointly captures short-run adjustment dynamics and the long-run equilibrium relationship:

$$\Delta \ln BREN T(t) = \alpha_0 + \lambda ECM(t-1) + \sum \beta_i \Delta \ln BREN T(t-i) + \sum \gamma_i \Delta GPR_POS(t-i) + \sum \delta_i \Delta GPR_NEG(t-i) + \sum \theta_i \Delta GPR_POS_RU(t-i) + \sum \phi_i \Delta GPR_NEG_RU(t-i) + \sum \mu_i \Delta GPR_POS_IR(t-i) + \sum \eta_i \Delta GPR_NEG_IR(t-i) + \varepsilon(t)$$

where ECM_{t-1} denotes the lagged long-run equilibrium error, λ represents the speed of adjustment, and ε_t is the error term. A negative and statistically significant λ indicates that short-run deviations are corrected toward the long-run equilibrium. Differences between the Russia–Ukraine and Iran interaction coefficients are evaluated using Wald tests.

4. Empirical Results

4.1. Unit root tests

Before model estimation, the stationarity properties of the variables are examined using ADF, Phillips–Perron, KPSS, and Zivot–Andrews unit root tests. The results are reported in Table 2.

Table 2. Results of ADF, PP, KPSS and Zivot–Andrews unit root tests

Variable	ADF	PP	KPSS	Zivot–Andrews	Break observation	Decision
LN_BRENT	-2.875*	-2.662	0.3264	-4.647	87	I(1)
LN_GPR	-3.760**	-5.177***	0.6362	-6.476***	174	I(0), with break
LN_EPUTRADE	-1.649	-3.139*	0.7150	-4.560	209	I(1)
LN_OVX	-4.3962***	-3.6183**	0.1424	-5.0074***	31	I(0)

Note: In the ADF and PP tests, *, ** and *** denote rejection of the unit root null at the 10%, 5% and 1% levels, respectively. In the KPSS test, the null hypothesis is stationarity. In the Zivot–Andrews test, the null hypothesis is a unit root without structural break.

The results indicate that the variables are integrated at either the I(0) or I(1) level, while none of the series is integrated of order two, I(2). This supports the use of the ARDL/NARDL bounds testing approach. The Zivot–Andrews results also highlight the importance of accounting for structural breaks in crisis-sensitive series such as geopolitical risk and oil market volatility.

4.2. Bounds test and error correction results

Table 3. Bounds test and error correction results for the comparative NARDL model

Panel	Model / Variable	Selected model	Criterion	Test statistic / Coefficient	Std. error	t-statistic	p-value	Decision
Bounds test	Comparative NARDL	ARDL(2,0,0,0,0,0)	SIC	5.069598	—	—	—	Cointegration at the 1% level
ECM	COINTEQ	—	—	-0.093492	0.015467	-6.031657	0.0000	Negative and significant

Note: Critical values: 10% $I(0)=2.120$, $I(1)=3.230$; 5% $I(0)=2.450$, $I(1)=3.610$; 1% $I(0)=3.150$, $I(1)=4.430$. A bounds F-statistic above the upper bound indicates cointegration. A negative and statistically significant ECM coefficient indicates adjustment toward long-run equilibrium.

The bounds test produces an F-statistic of 5.069598, which exceeds the upper-bound critical value of 4.430 at the 1% significance level. Therefore, the null hypothesis of no cointegration is rejected. This indicates a strong long-run relationship between Brent oil prices, the general GPR components, and the event-contextual interaction terms.

The error correction coefficient is -0.093492 and statistically significant at the 1% level. Its negative sign indicates that the long-run equilibrium mechanism operates properly. Given the monthly frequency of the data, approximately 9.35% of short-run disequilibria are corrected in the following month.

4.3. Long-run coefficients

Table 4. Long-run coefficients of the comparative NARDL model

Variable	Coefficient	Std. error	t-statistic	p-value	Result
GPR_POS	-0.015172	0.006482	-2.347805	0.0198	Significant
GPR_NEG	-0.014896	0.006505	-2.289737	0.0230	Significant
GPR_POS_RU	0.028413	0.010579	2.685710	0.0070	Strongly significant
GPR_NEG_RU	0.028189	0.010607	2.657643	0.0086	Strongly significant
GPR_POS_IR	0.011631	0.006961	1.670993	0.0962	Significant at 10%
GPR_NEG_IR	0.011844	0.006958	1.702263	0.0901	Significant at 10%

Note: GPR_POS and GPR_NEG represent the general geopolitical risk components. GPR_POS_RU and GPR_NEG_RU represent Russia–Ukraine interactions, while GPR_POS_IR and GPR_NEG_IR represent Iran-related interactions.

The general GPR components, GPR_POS and GPR_NEG, are negative and statistically significant. This suggests that when event context is not taken into account, the general effect of geopolitical risk on Brent prices is negative. This is consistent with the view that geopolitical risk may depress oil prices through weaker economic activity and lower demand expectations, in addition to supply disruption channels.

In contrast, the Russia–Ukraine interaction variables are positive and strongly significant. The coefficients of GPR_POS_RU and GPR_NEG_RU are 0.028413 and 0.028189, respectively, and both are significant at the 1% level. These results suggest that geopolitical risk transmission to Brent prices strengthened during the Russia–Ukraine War, likely through sanctions, producer-country risk, market expectations, and European energy security concerns.

Iran-related interaction variables are also positive, but their significance is weaker. GPR_POS_IR and GPR_NEG_IR are significant only at the 10% level. This indicates that Iran-related tensions affect Brent prices, but the effect is less statistically pronounced than the Russia–Ukraine effect and appears to operate mainly through the oil risk premium channel associated with the Strait of Hormuz, transit security, tanker transportation, and expected supply disruptions.

4.4. Event-contextual comparison: Wald tests

Table 5. Comparison of Russia–Ukraine and Iran interaction coefficients

Test	Hypothesis	Restriction	F-statistic	p-value	Decision
Difference in positive GPR effect	Positive GPR effects for Russia–Ukraine and Iran are equal	$C(5)=C(7)$	1.862110	≈ 0.173	Difference is not significant
Difference in negative GPR effect	Negative GPR effects for Russia–Ukraine and Iran are equal	$C(6)=C(8)$	1.793035	≈ 0.181	Difference is not significant

Note: The null hypothesis is that the relevant Russia–Ukraine and Iran interaction coefficients are equal. A p-value above 0.10 indicates that the coefficient difference is not statistically significant.

The Wald tests show that the differences between the Russia–Ukraine and Iran interaction coefficients are not statistically significant. Although the Russia–Ukraine coefficients are larger than the Iran-related coefficients, this difference is not statistically confirmed. The results therefore define the interpretive boundary of the study: coefficient magnitudes suggest event-contextual differentiation, but Wald tests do not confirm the difference statistically.

4.5. Diagnostic and robustness checks

Table 6. Diagnostic test results for the comparative NARDL model

Test	Test statistic / Finding	p-value	Decision
Breusch–Godfrey LM	$F \approx 0.001183$	≈ 0.9988	No serial correlation
Breusch–Pagan–Godfrey	$F \approx 1.709704$	≈ 0.0974	No heteroskedasticity at 5%; borderline at 10%
Jarque–Bera	839.1477	0.0000	Residuals are not normally distributed
Ramsey RESET	$F = 6.353562$	0.012443	Functional form/specification concern
CUSUM	Within the critical bounds	—	Parameter stability supported
CUSUMSQ	Generally within the critical bounds	—	Overall stability supported; final-period movement toward boundary

Note: The Breusch–Godfrey LM test evaluates serial correlation; Breusch–Pagan–Godfrey evaluates heteroskedasticity; Jarque–Bera evaluates residual normality; Ramsey RESET evaluates functional form; CUSUM and CUSUMSQ evaluate parameter stability.

The diagnostic tests indicate that the model performs well in terms of serial correlation and parameter stability. The Jarque–Bera test rejects normality, which is not unexpected in oil price series affected by geopolitical crises and extreme observations. The Ramsey RESET result signals a possible functional form or specification concern. For this reason, robustness checks are used to assess whether the main findings are preserved across alternative specifications.

Table 7. Summary of robustness checks

Robustness model	Model modification	Bounds F-statistic	Decision
Alternative model 1	New model specification	4.303463	Cointegration exists
Alternative model 2	Alternative lag / model structure	5.055999	Cointegration exists
OVX-controlled model	Oil market volatility control added	9.214597	Strong cointegration
EPUTRADE-controlled model	Trade policy uncertainty control added	4.544833	Cointegration exists
Reduced model	Model simplified	8.409385	Strong cointegration
Russia–Ukraine sub-model	Only Russia–Ukraine interactions included	6.066879	Cointegration exists
Iran sub-model	Only Iran interactions included	4.645979	Cointegration exists

Note: The bounds F-statistics indicate whether the long-run relationship is preserved under alternative specifications. A statistic above the relevant upper-bound critical value indicates cointegration.

The robustness checks show that the main findings are generally preserved across alternative model specifications. The OVX-controlled model produces a particularly high F-statistic, indicating that the long-run relationship remains strong even after controlling for oil market volatility. The EPUTRADE-controlled model also preserves the cointegration relationship, suggesting that the findings are not solely explained by trade policy uncertainty. Separate Russia–Ukraine and Iran sub-models further indicate that both event contexts constitute relevant geopolitical risk transmission channels for Brent oil prices.

5. Conclusion and Policy Implications

This study shows that the impact of geopolitical risk on Brent oil prices cannot be explained solely by the general level of risk; rather, the event context in which risk emerges plays a decisive role in the pricing process. The empirical findings indicate that the event-contextual model, which incorporates Russia–Ukraine War and Iran-related tension interactions, reveals a strong long-run relationship. Geopolitical risk in oil markets should therefore not be treated as a homogeneous source of uncertainty, but as a pricing mechanism that varies according to the nature of the event, its geography, and the position of risk-generating actors within the energy supply chain.

The findings suggest that geopolitical risk transmission to Brent prices is stronger and positive during the Russia–Ukraine War period. This reflects the role of sanctions, producer-country risk, European energy security concerns, and expectations of supply disruptions. Iran-related tensions also generate a positive but weaker effect, mainly through the Strait of Hormuz, tanker transportation, transit security, and the oil risk premium channel. However, the Wald tests show that the coefficient difference between the two event contexts is not statistically conclusive. The Russia–Ukraine effect therefore appears stronger, but this difference should be interpreted cautiously.

These findings have direct implications for energy security policy. Geopolitical risk monitoring should not rely only on aggregate risk indicators; it should also track production geographies, sanctions regimes, energy corridors, chokepoints, and maritime transport routes. Producer-country crises and transit-security crises require different policy responses, including differentiated early-warning systems, flexible strategic reserve management, supply diversification, and coordinated monitoring of chokepoint vulnerability. In this sense, geopolitical risk is not only an economic uncertainty shock in Brent oil markets, but also a spatially embedded and context-specific pricing mechanism.

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